

Unit-3 System of linear algebraic equations

Existence of Solution

Consider the system of equations :-

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$$\vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots$$

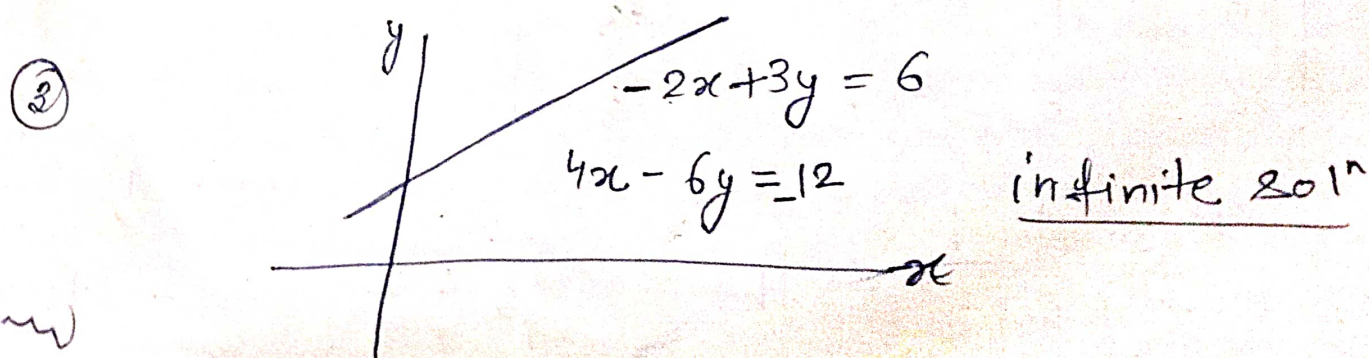
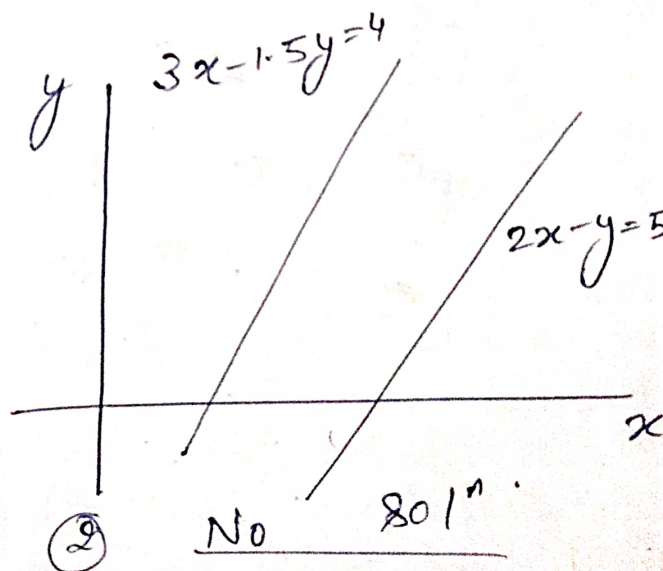
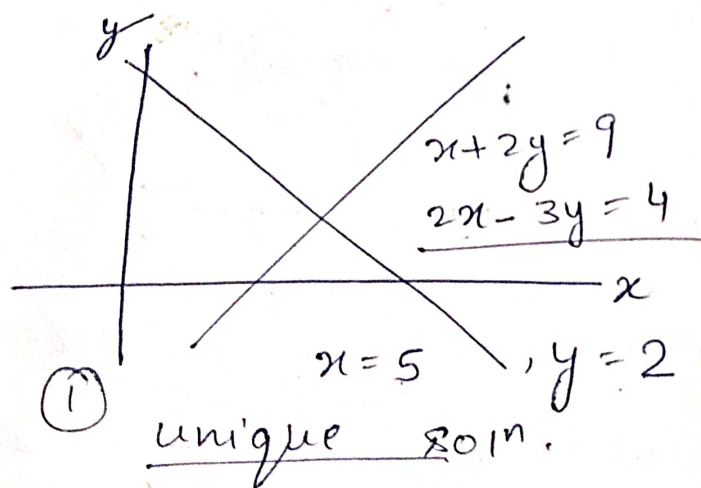
$$a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n = b_n$$

Then, there are following possibilities.

1) System has unique soln.

2) System has no solution.

3) System has infinite soln.



Gauss elimination method

Here we solve the system of equations by triangularisation.

Eg

$$\begin{aligned} 3x_1 + 6x_2 + x_3 &= 16 \\ 2x_1 + 4x_2 + 3x_3 &= 13 \\ x_1 + 3x_2 + 2x_3 &= 9 \end{aligned}$$

Matrix corresponding to the system,

$$[A:b] \sim \left[\begin{array}{ccc|c} 3 & 6 & 1 & 16 \\ 2 & 4 & 3 & 13 \\ 1 & 3 & 2 & 9 \end{array} \right] \rightarrow \text{Augmented matrix}$$

$$\left[\begin{array}{ccc|c} 3 & 6 & 1 & 16 \\ 0 & 0 & 7/3 & 7/3 \\ 0 & 1 & 5/3 & 11/3 \end{array} \right] \quad \begin{aligned} R_2 &\rightarrow R_2 - \frac{2}{3} R_1 \\ R_3 &\rightarrow R_3 - \frac{1}{3} R_1 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 3 & 6 & 1 & 16 \\ 0 & 1 & 5/3 & 11/3 \\ 0 & 0 & 7/3 & 7/3 \end{array} \right] \quad R_2 \leftrightarrow R_3$$

$$7x_3 = 7 \Rightarrow x_3 = 1$$

$$x_1 = \frac{16 - 12 - 1}{3}$$

$$3x_2 + 5x_3 = 11 \Rightarrow x_2 = 2, \quad = 1$$

Computational efforts in Gauss elimination method

Here we estimate the computational effort required in terms of arithmetic operations

	Elimination process	Substitution process	Both stages
• Multiplication	$\frac{n}{3} (n^2 - 1)$	$\frac{n}{2} (n - 1)$	$\frac{(n-1)n(2n+5)}{6}$
• Subtraction	$\frac{n}{3} (n^2 - 1)$	$\frac{n}{2} (n - 1)$	$\frac{(n-1)n(2n+5)}{6}$
• Division	$\frac{n(n^2 - 1)}{2}$	n	$\frac{n(n+1)}{2}$

Eg:-

$$\begin{aligned} 2x_1 + x_2 + x_3 &= 10 \\ 3x_1 + 2x_2 + 3x_3 &= 18 \\ x_1 + 4x_2 + 9x_3 &= 16 \end{aligned}$$

$$[A:b] \sim \left[\begin{array}{ccc|c} 2 & 1 & 1 & 10 \\ 3 & 2 & 3 & 18 \\ 1 & 4 & 9 & 16 \end{array} \right]$$

$$\begin{aligned} R_2 &\rightarrow R_2 - \frac{3}{2} R_1 \\ R_3 &\rightarrow R_3 - \frac{1}{2} R_1 \end{aligned}$$

$$\sim \left[\begin{array}{ccc|c} 2 & 1 & 1 & 10 \\ 0 & 1/2 & 3/2 & 3 \\ 0 & 7/2 & 17/2 & 11 \end{array} \right]$$

$$R_3 \rightarrow R_3 - 7R_2$$

$$\sim \left[\begin{array}{ccc|c} (2) & 1 & 1 & 10 \\ 0 & (1/2) & 3/2 & 3 \\ 0 & 0 & (-2) & -10 \end{array} \right]$$

$$\begin{aligned} 2x_3 &= 10 \Rightarrow x_3 = 5 \\ x_2 &= 6 - 3x_3 = -9 \\ x_1 &= \frac{10 + 9 - 5}{2} = 7 \end{aligned}$$

↪ Chaired elements are known as pivot elem.

Ex:- Solve the following by Gauss-elimination method.

(a)
$$\begin{aligned} x + y + z &= 3 \\ 2x + 3y + 3z &= 10 \\ 3x - y + 2z &= 13 \end{aligned}$$

$$(-5, 8/6, 16/3)$$

(b)
$$\begin{aligned} 2x + y + 4z &= 12 \\ 8x - 3y + 2z &= 23 \\ 4x + 11y - z &= 33 \end{aligned}$$

$$(0.975, 2.81, 1.81)$$

(2) Gauss Jordan elimination method :-

Ex.
$$[A:b] \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 2 & -3 & 4 & 13 \\ 3 & 4 & 5 & 40 \end{array} \right]$$

$$x + y + z = 9$$

$$2x - 3y + 4z = 13$$

$$3x + 4y + 5z = 40$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 0 & -5 & 2 & -5 \\ 0 & 1 & 2 & 13 \end{array} \right]$$

$$R_2 \rightarrow R_2 - 2R_1$$

$$R_3 \rightarrow R_3 - 3R_1$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 0 & 1 & -2/5 & 1 \\ 0 & 1 & 2 & 13 \end{array} \right]$$

$$R_2 \rightarrow R_2 / (-5)$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 7/5 & 8 \\ 0 & 1 & -2/5 & 1 \\ 0 & 0 & 12/5 & 12 \end{array} \right]$$

$$R_1 \rightarrow R_1 - R_2$$

$$R_3 \rightarrow R_3 - R_2$$

$$\begin{bmatrix} 1 & 0 & 7/5 & 8 \\ 0 & 1 & -2/5 & 1 \\ 0 & 0 & 1 & \frac{5}{2} \end{bmatrix}$$

$$R_3 \rightarrow 5R_3 / 8$$

$$\begin{bmatrix} 1 & 0 & 0 & \frac{11}{2} \\ 0 & 1 & 0 & 9 \\ 0 & 0 & 1 & \frac{45}{2} \end{bmatrix}$$

$$R_2 \rightarrow R_2 + \frac{2}{5} R_3$$

$$R_1 \rightarrow R_1 - \frac{7}{5} R_3$$

$$x_1 = \frac{11}{2}, x_2 = 9, x_3 = \frac{45}{2}$$

Ex: a) $x + 2y + 2z = 3$
 $2x + 3y + 3z = 10$
 $3x - y + 2z = 13$
 $(-2, 1, 3)$

b) $10x + y + 2z = 13$
 $2x + 10y + 2z = 10$
 $x + y + 5z = 7$
 $(1, 1, 1)$

Computational efforts

Total operations required to complete the process

Multiplication :- $\frac{1}{2} n(n^2 - 1)$

Subtraction :- $\frac{n(n^2 + 1)}{2}$

Divisions :- $\frac{n(n-1)}{2}$

Triangular factorization methods

Factorize A into ~~two~~ $L \cdot U$.

i.e., $A = L \cdot U$, where $L =$ lower Δ^l matrix
 $U =$ upper Δ^u matrix

i.e.,
$$L = \begin{bmatrix} d_{11} & 0 & 0 & \dots & 0 \\ d_{21} & d_{22} & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ d_{n1} & d_{n2} & \dots & \dots & d_{nn} \end{bmatrix}_{n \times n}$$

$$U = \begin{bmatrix} u_{11} & u_{12} & \dots & u_{1n} \\ 0 & u_{22} & \dots & u_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & u_{nn} \end{bmatrix}_{n \times n}$$

$$\therefore AX = b \Rightarrow (LU)X = b \Rightarrow L(UX) = b$$

Let $UX = z$, for unknown z .

then we get $Lz = b$.

Next solve $Lz = b$ to find z

then solve $UX = z$ to find X ,
which is the desired solⁿ of $AX = b$.

① Doolittle Method :-

Here we assume the diagonal entries of L to be unity.

i.e., let
$$L = \begin{bmatrix} 1 & \textcircled{d_{12}} & \textcircled{d_{13}} \\ d_{21} & 1 & \textcircled{d_{23}} \\ d_{31} & d_{32} & 1 \end{bmatrix}, U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

Eg Solve the system

$$\begin{aligned} 3x_1 + 2x_2 + x_3 &= 10 \\ 2x_1 + 3x_2 + 2x_3 &= 14 \\ x_1 + 2x_2 + 3x_3 &= 14 \end{aligned}$$

by Little LU decomposition method.

Soln let $L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix}$

$$U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

$$A = LU \Rightarrow \begin{bmatrix} 3 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 3 \end{bmatrix} =$$

$$\begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix}$$

$$\Rightarrow u_{11} = 3,$$

$$u_{12} = 2$$

$$u_{13} = 1$$

$$l_{21} = 2/3$$

$$l_{31} = 1/3$$

$$l_{32} = \left(2 - \frac{2}{3}\right) / \frac{5}{3} = \frac{4}{5}$$

$$u_{22} = 3 - \frac{4}{3} = \frac{5}{3}$$

$$u_{23} = 2 - \frac{2}{3} = \frac{4}{3}, \quad u_{33} = 3 - \frac{1}{3} - \frac{16}{15} = \frac{24}{15} = \frac{8}{5}$$

$$\therefore A = LU$$

$$\Rightarrow \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix}$$

$$\Rightarrow \begin{array}{l|l} u_{11} = a_{11} & l_{21} = a_{21}/u_{11} \\ u_{12} = a_{12} & l_{31} = a_{31}/u_{11} \\ u_{13} = a_{13} & l_{32} = \frac{a_{32} - l_{31}u_{12}}{u_{22}} \\ u_{22} = a_{22} - l_{21}u_{12} & \\ u_{23} = a_{23} - l_{21}u_{13} & \\ u_{33} = a_{33} - l_{31}u_{13} - l_{32}u_{23} & \end{array}$$

In this way, we get the entries of L & U . Now solve $UX = b$ to get X .
then solve $UX = z$ to find X .

$$\therefore L = \begin{bmatrix} 1 & 0 & 0 \\ 2/3 & 1 & 0 \\ 1/3 & 4/5 & 1 \end{bmatrix}, U = \begin{bmatrix} 3 & 2 & 1 \\ 0 & 5/3 & 4/3 \\ 0 & 0 & 8/5 \end{bmatrix}$$

$$Lz = b$$

$$\Rightarrow z_1 = 10$$

$$\frac{2}{3}z_1 + z_2 = 14$$

$$\Rightarrow z_2 = 14 - \frac{20}{3} = \frac{22}{3}$$

$$z_3 = 14 - \frac{10}{3} - \frac{88}{15}$$

$$= \frac{72}{15}$$

$$Ux = z$$

$$\Rightarrow \frac{8x_3}{5} = z_3 = \frac{72}{15}$$

$$\Rightarrow x_3 = 3$$

$$x_2 = \frac{3}{5} \left(\frac{22}{3} - 4 \right) = 2$$

$$x_1 = \frac{10 - 4 - 3}{3} = 1$$

$$\therefore (1, 2, 3) \text{ Ans.}$$

② Crout's method

In this method we assume diagonal entries of U as 1 then proceed same as Doolittle method.

$$\text{i.e., } L = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}, U = \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

Eg. Solve by Crout's method.

$$2x + y + 4z = 12$$

$$8x - 3y + 2z = 20$$

$$4x + 11y - 2z = 33$$

Ex 11 $A = \begin{bmatrix} 2 & 1 & 4 \\ 8 & -3 & 2 \\ 4 & 11 & -1 \end{bmatrix}$

let $L = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}$, $U = \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}$

$A = LU$

$$\Rightarrow \begin{bmatrix} 2 & 1 & 4 \\ 8 & -3 & 2 \\ 4 & 11 & -1 \end{bmatrix} = \begin{bmatrix} l_{11} & l_{11}u_{12} & l_{11}u_{13} \\ l_{21} & l_{21}u_{12} + l_{22} & l_{21}u_{13} + l_{22}u_{23} \\ l_{31} & l_{31}u_{12} + l_{32} & l_{31}u_{13} + l_{32}u_{23} + l_{33} \end{bmatrix}$$

$\Rightarrow l_{11} = 2, l_{21} = 8, l_{31} = 4, l_{22} = -7$

$u_{12} = 1/2, u_{13} = 2, u_{23} = \frac{2-16}{-7} = 2$

$l_{32} = 11 - 2 = 9, l_{33} = -1 - 8 - 18 = -27$

$Lz = b \Rightarrow \begin{bmatrix} 2 & 0 & 0 \\ 8 & -7 & 0 \\ 4 & 9 & -27 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} 12 \\ 20 \\ 33 \end{bmatrix}$

$z_1 = 6$

$z_2 = \frac{20 - 8(6)}{-7} = \frac{-28}{-7} = 4$

$z_3 = \frac{33 - 24 - 36}{-27} = \frac{-27}{-27} = 1$

$$UX = X$$

$$\Rightarrow \begin{bmatrix} 1 & 1/2 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ 1 \end{bmatrix}$$

$$x_3 = 1, \quad x_2 = 4 - 2 = 2,$$

$$x_1 = 6 - 1 - 2 = 3$$

Ex solve by Crout's method!

$$2x_1 + x_2 + x_3 = 7$$

$$x_1 + 2x_2 + x_3 = 8$$

$$x_1 + x_2 + 2x_3 = 9$$

Ans.
(1, 2, 3)

③ Cholesky method (or square root method)

This method is applicable for symmetric matrix A

$$A = LU \quad \& \quad A^t = A$$

$$\Rightarrow (LU)^t = LU$$

$$\Rightarrow U^t L^t = LU \Rightarrow L = U^t \quad \& \quad U = L^t$$

$$\therefore A = U^t U = LL^t$$

$$\Rightarrow LL^t X = b \quad (\text{for solving } AX = b)$$

$$\Rightarrow L(L^t X) = b \quad \therefore \text{let } L^t X = Z$$

$$\Rightarrow \text{Solve } LZ = b \quad \text{then obtain } X \text{ to find } L^t X = Z.$$

Q. Apply Cholesky method to solve,

$$25x + 15y - 5z = 35$$

$$15x + 18y + 0z = 33$$

$$-5x + 0y + 11z = 6$$

Solⁿ $\Rightarrow$

$$A = \begin{bmatrix} 25 & 15 & -5 \\ 15 & 18 & 0 \\ -5 & 0 & 11 \end{bmatrix}$$

$\Rightarrow A$ is symmetric.

$$A = LL^T$$

$$\Rightarrow A = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} l_{11} & l_{21} & l_{31} \\ 0 & l_{22} & l_{32} \\ 0 & 0 & l_{33} \end{bmatrix}$$

$$= \begin{bmatrix} l_{11}^2 & l_{11}l_{21} & l_{11}l_{31} \\ l_{21}l_{11} & l_{21}^2 + l_{22}^2 & l_{21}l_{31} + l_{22}l_{32} \\ l_{31}l_{11} & l_{31}l_{21} + l_{32}l_{22} & l_{31}^2 + l_{32}^2 + l_{33}^2 \end{bmatrix}$$

$$\Rightarrow \begin{array}{l|l} l_{11}^2 = 25, & l_{11} = 5 \\ l_{21} = 15/5 = 3 & \\ l_{31} = -5/5 = -1 & \\ l_{22} = \sqrt{18 - 9} = 3 & \\ l_{32} = \frac{+3}{3} = 1 & \end{array} \quad \begin{array}{l} l_{33}^2 = \\ 11 - 1 - 1 = 9 \\ l_{33} = 3 \end{array}$$

$$\Rightarrow L = \begin{bmatrix} 5 & 0 & 0 \\ 3 & 3 & 0 \\ -1 & 1 & 3 \end{bmatrix}$$

$$Lz = b \Rightarrow 5z_1 = 35, z_1 = 7$$

$$3z_1 + 3z_2 = 33$$

$$\Rightarrow z_2 = \frac{33 - 21}{3} = 4$$

$$3z_3 = 6 + 7 - 4 = 9$$

$$\Rightarrow z_3 = 3$$

$$L^t x = z \Rightarrow \begin{bmatrix} 5 & 3 & -1 \\ 0 & 3 & 1 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ 3 \end{bmatrix}$$

$$\Rightarrow x_3 = 1$$

$$x_2 = 1, x_1 = 5/5 = 1 \quad \underline{\text{Ans}}$$

Ex: Solve by Cholesky's method,

$$10x + y + z = 12$$

$$x + 10y + z = 12$$

$$x + y + 10z = 12$$

$(1, 1, 1)$ Ans.

Eigen value problem

(Power method or Rayleigh's method)

Using this method, we find the dominant eigen-value (largest eigenvalue)

- ① start with an initial approximation to the eigen vector say X .
- ② Evaluate AX which is written as $\lambda^{(1)} X^{(1)}$; This gives first approximation $\lambda^{(1)}$ to eigen-value & $X^{(1)}$ to eigen-vector.
- ③ Similarly evaluate $AX^{(1)} = \lambda^{(2)} X^{(2)}$ which gives 2nd approximation $\lambda^{(2)}$ to eigen-value & $X^{(2)}$ to eigen-vector.
- ④ Repeat the process till $[X^{(n)} - X^{(n-1)}]$ becomes negligible.
- ⑤ Then $\lambda^{(n)}$ will be the dominant or required eigen-value & $X^{(n)}$ will be the corresponding eigen-vector.

Ex 9 Determine the largest eigen-value & corresponding eigen-vector of the matrix $\begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$

Solⁿ let initial approximation be $X = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

then $AX = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix} = 5 \begin{bmatrix} 1 \\ 0.2 \end{bmatrix}$
 $= \lambda^{(1)} \cdot X^{(1)}$

$\Rightarrow$ 1st appⁿ to eigen value is $\lambda^{(1)} = 5$
& eigenvector $X^{(1)} = \begin{bmatrix} 1 \\ 0.2 \end{bmatrix}$.

Next $AX^{(1)} = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0.2 \end{bmatrix} = \begin{bmatrix} 5.8 \\ 1.4 \end{bmatrix} = 5.8 \begin{bmatrix} 1 \\ .241 \end{bmatrix}$
 $= \lambda^{(2)} X^{(2)}$

$\Rightarrow$ 2nd appⁿ to eigen value is $\lambda^{(2)} = 5.8$
& $X^{(2)} = \begin{bmatrix} 1 \\ 0.241 \end{bmatrix}$

Continuing like this,

$$AX^{(2)} = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ .241 \end{bmatrix} = \begin{bmatrix} 5.966 \\ 1.482 \end{bmatrix}$$

$$= 5.966 \begin{bmatrix} 1 \\ .248 \end{bmatrix} = \lambda^{(3)} \cdot X^{(3)}$$

3rd $\rightarrow \lambda^{(3)} = 5.966$, $X^{(3)} = \begin{bmatrix} 1 \\ .248 \end{bmatrix}$

$$AX^{(3)} = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ .248 \end{bmatrix} = \begin{bmatrix} 5.994 \\ 1.496 \end{bmatrix} = 5.994 \begin{bmatrix} 1 \\ .250 \end{bmatrix}$$

$$4^{th} \rightarrow \lambda^{(4)} = 5.994, \quad X^{(4)} = \begin{bmatrix} 1 \\ .250 \end{bmatrix}$$

$$5^{th} \rightarrow AX^{(4)} = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ .250 \end{bmatrix} = 5.999 \begin{bmatrix} 1 \\ .25 \end{bmatrix}$$

$$\Rightarrow \lambda^{(5)} = 5.999, \quad X^{(5)} = \begin{bmatrix} 1 \\ .25 \end{bmatrix}$$

$$AX^{(5)} = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0.25 \end{bmatrix} = 6 \begin{bmatrix} 1 \\ 0.25 \end{bmatrix} = \lambda^{(6)} X^{(6)}$$

$$6^{th} \rightarrow \lambda^{(6)} = 6, \quad X^{(6)} = X^{(5)} = \begin{bmatrix} 1 \\ .25 \end{bmatrix}$$

So $[X^{(6)} - X^{(5)}]$ is negligible.

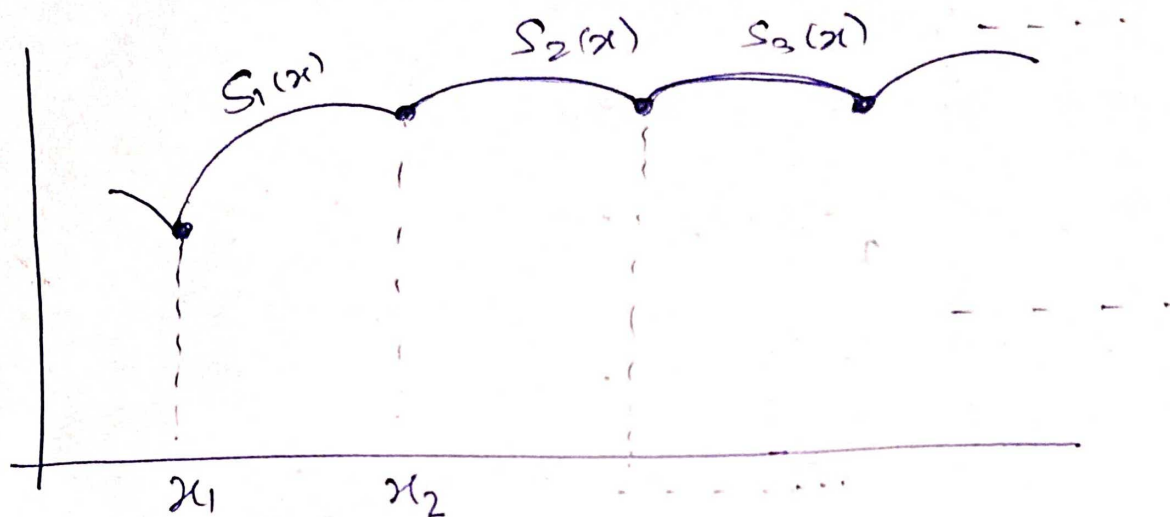
$\therefore \lambda^{(6)} = 6$ is dominant eigenvalue & largest of all, $X^{(5)}$ is corresponding eigen-vector.

Eg Repeat above question for $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$
Take $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ as initial eigen-vector.

Approximation by Spline functions

Consider the interval $[a, b]$, it is divided into n sub-intervals $[x_0, x_1], [x_1, x_2], \dots, [x_{n-1}, x_n]$ where
$$a = x_0 < x_1 < x_2 < \dots < x_{n-1} < x_n = b$$

interval. nodes



The polynomials $S_i(x)$ that preserve the discontinuities at connecting points $x_1, \dots, x_{n-1}$ are called spline functions.

Properties

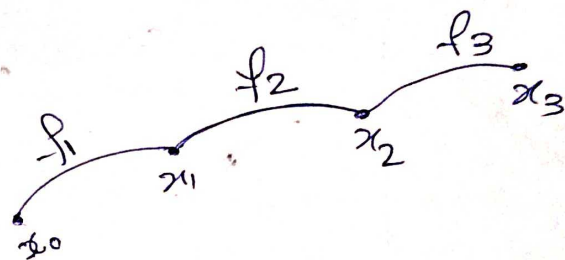
- ① $S(x)$ is a polynomial of deg. m in each sub-interval $[x_i, x_{i+1}]$, $i = 0, 1, \dots, n$.
- ② $S(x)$ & its derivatives of order $1, 2, \dots, m-1$ are continuous in the range $[x_0, x_n]$.

eg Check whether following polynomials are splines or not.

(i) $f(x) = \begin{cases} x+1 & -1 \leq x \leq 0 \\ 2x+1 & 0 \leq x \leq 1 \\ 4-x & 1 \leq x \leq 2 \end{cases}$

Here $x_0 = -1, x_1 = 0, x_2 = 1, x_3 = 2$
 $n = 4$, $\underbrace{x_1, x_2}_{\text{internal nodes knots}}$

$$\begin{aligned} f_1(x) &= x+1 \\ f_2(x) &= 2x+1 \\ f_3(x) &= 4-x \end{aligned}$$



$$f_1(x_1) = 1 = f_2(x_1) = 1$$

$$f_2(x_2) = 3 = f_3(x_2) = 3$$

since each f_i is continuous $\therefore f(x)$ is linear spline, but, f_i' is not continuous

as! $f_1' = 1$ gives different values
 $f_2' = 2$ at each internal nodes
 $f_3' = -1$ $\therefore f$ is not 2nd deg. Spline.

Note If f is continuous but not its 1st order derivative then f is s.t.b. linear spline but not 2nd deg. spline.

(ii) $f(x) = \begin{cases} x & 0 \leq x \leq 1 \\ x^2 - x + 1 & 1 \leq x \leq 2 \\ 3x - 3 & 2 \leq x \leq 3 \end{cases}$

Eg
=

Given

x	0	1	2	3
$f(x)$	1	2	33	244

Fit a quadratic splines, given $f''(0) = 0$.
Estimate $f(2.5)$.

Let the splines be

$$f_1(x) = a_1x^2 + b_1x + c_1 \quad 0 \leq x \leq 1$$

$$f_2(x) = a_2x^2 + b_2x + c_2 \quad 1 \leq x \leq 2$$

$$f_3(x) = a_3x^2 + b_3x + c_3 \quad 2 \leq x \leq 3$$

$$f''(0) \Rightarrow a_1 = 0$$

$$x_0 = 0 \Rightarrow f(0) = 1 \Rightarrow c_1 = 1$$

$$x_1 = 1 \Rightarrow f(1) = 2 \Rightarrow a_1 + b_1 = 1 \Rightarrow b_1 = 1$$

$$a_2 + b_2 + c_2 = 2 \quad \text{--- (1)}$$

$$x_2 = 2 \Rightarrow f(2) = 33 \Rightarrow 4a_2 + 2b_2 + c_2 = 33 \quad \text{--- (2)}$$

$$4a_3 + 2b_3 + c_3 = 33 \quad \text{--- (4)}$$

$$x_3 = 3 \Rightarrow f(3) = 244 \Rightarrow 9a_3 + 3b_3 + c_3 = 244 \quad \text{--- (5)}$$

Also $f_1(x_1) = f_2(x_1)$

$$a_1 + b_1 + c_1 = a_2 + b_2 + c_2$$

$$\Rightarrow f_1'(x_1) = f_2'(x_1)$$

2nd deg. spline $\Rightarrow 2a_1 + b_1 = 2a_2 + b_2$

$$\text{--- (3)}$$

By ① ② ③,

$$a_2 = 30$$

$$b_2 = -59$$

$$c_2 = 31$$

Solⁿ
 $x_0 = 0, x_1 = 1, x_2 = 2, x_3 = 3$

$$f_1(x) = x$$

$$f_1'(x) = 1$$

$$f_2(x) = x^2 - x + 1$$

$$f_2'(x) = 2x - 1$$

$$f_3(x) = 3x - 3$$

$$f_3'(x) = 3$$

$$f_1(1) = 1 = f_2(1)$$

$$f_2(2) = 3 = f_3(2)$$

$\Rightarrow f$ is linear spline

$$f_1'(1) = 1 = f_2'(1)$$

$$f_2'(2) = 3 = f_3'(2)$$

$\Rightarrow f$ is quadratic spline

Ex¹ Repeat above question for,

$$f(x) = \begin{cases} x^2 + 1 & 0 \leq x \leq 1 \\ 2x^2 & 1 \leq x \leq 2 \\ 5x - 2 & 2 \leq x \leq 3 \end{cases}$$

Note Each interval has a different polynomial of deg. m or less. The set of all polynomials form a spline interpolation polynomial. The process of constructing such polynomials is known as spline interpolation.

$$f_2'(2) = f_3'(2)$$

$$\Rightarrow \cancel{4a_2 + 2b_2 + c_2 = 0}$$

$$4a_2 + b_2 = 4a_3 + b_3$$

$$\Rightarrow 4a_3 + b_3 = 120 - 59 = 61 \quad - (6)$$

Using (4), (5), (6)

$$a_3 = 150, \quad b_3 = -539, \quad c_3 = 511$$

$$f(x) = \begin{cases} x+1, & 0 \leq x < 1 \\ 30x^2 - 59x + 31, & 1 \leq x < 2 \\ 150x^2 - 539x + 511, & 2 \leq x < 3 \end{cases}$$

$$\begin{aligned} f(2.5) &= 150(2.5)^2 - 539(2.5) + 511 \\ &= 101. \end{aligned}$$

Cubic splines Case (i) if knots are not equi-spaced.

$$S_i(x) = \frac{a_{i-1}}{6h_i} (h_i^2 u_i - u_i^3) + \frac{a_i}{6h_i} (u_i^3 - h_i^2 u_{i-1}) + \frac{1}{h_i} (f_i u_{i-1} - f_{i-1} u_i) \quad - (1)$$

where $u_i = x - x_i$
 $h_i = x_i - x_{i-1}$

Also,

$$h_i a_{i-1} + 2(h_i + h_{i+1}) a_i + h_{i+1} a_{i+1} = 6 \left(\frac{f_{i+1} - f_i}{h_{i+1}} - \frac{f_i - f_{i-1}}{h_i} \right) \quad - (2)$$

Eqn ② when written for $i=1, \dots, n-1$,
 we get $n-1$ equations containing $n+1$
 unknowns $a_0, a_1, \dots, a_n$.

Applying the condition that derivatives at
 the end pts. are zero, we get
 $a_0 = a_n = 0$.

Ex Given

i	0	1	2
x_i	4 (x_0)	9 (x_1)	16 (x_2)
f_i	2	3	4

estimate the function value at $x=7$,
 using Cubic splines.

Solⁿ

$$h_1 = x_1 - x_0 = 9 - 4 = 5$$

$$h_2 = x_2 - x_1 = 16 - 9 = 7$$

$$f_0 = 2, f_1 = 3, f_2 = 4.$$

Since $a_0 = a_2 = 0$

Using ② for $i=1$,

$$h_1 a_0 + 2(h_1 + h_2)a_1 + h_2 a_2 = 6 \left(\frac{f_2 - f_1}{h_2} - \frac{f_1 - f_0}{h_1} \right)$$

$$\Rightarrow 2(12) \cdot a_1 + \cancel{7a_2} = 6 \left(\frac{(4-3)}{7} - \frac{(3-2)}{5} \right)$$

$$\Rightarrow 24a_1 = \frac{6(-2)}{35} \Rightarrow a_1 = \frac{-12}{35(24)} = \underline{\underline{-0.0143}}$$

Since $n=3 \therefore \exists$ 2 Cubic splines as,

$$S_1(x), \quad x_0 \leq x \leq x_1$$

$$S_2(x), \quad x_1 \leq x \leq x_2$$

$$\text{i.e., } f(x) = \begin{cases} S_1(x), & 4 \leq x \leq 9 \\ S_2(x), & 9 \leq x \leq 16 \end{cases}$$

To find $f(7)$, we only need to find only $S_1(x)$,

$$\text{by ① } S_1(x) = \frac{a_0}{6h_1} (h_1^2 u_1 - u_1^3) + \frac{a_1}{6h_1} (u_0^3 - h_1^2 u_0)$$

$$+ \frac{1}{h_1} (f_1 u_0 - f_0 u_1)$$

$$= \frac{a_1 (u_0^3 - h_1^2 u_0)}{6h_1} + \frac{1}{h_1} (f_1 u_0 - f_0 u_1)$$

$$= \frac{(-0.0143) (u_0^3 - 25u_0)}{30} + \frac{1}{5} (3u_0 - 2u_1)$$

$$u_0 = x - x_0, \quad u_1 = x - x_1$$

$$= 7 - 4 = 3, \quad = 7 - 9 = -2$$

$$\therefore S_1(7) = \frac{(-0.0143) (27 - 75)}{30} + \frac{1}{5} (9 + 4)$$

$$= \underline{\underline{2.6229}}$$

case (ii) If knots are ~~not~~ equi-spaced

The general system of $n-1$ equations contained in eqn (2) can be expressed as,

$$\begin{bmatrix} 2(h_1+h_2) & h_2 & 0 & \dots & 0 & 0 & \dots & 0 \\ h_2 & 2(h_2+h_3) & h_3 & \dots & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & h_{n-2} & 2(h_{n-2}+h_{n-1}) & h_{n-1} & 0 \\ & & & & 0 & h_{n-1} & 2(h_{n-1}+h_n) & h_n \end{bmatrix}$$

$$\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_{n-1} \end{bmatrix} = \begin{bmatrix} D_1 \\ D_2 \\ \vdots \\ D_{n-1} \end{bmatrix} \quad \text{--- (3)}$$

If points are equispaced then, substitute $h_1 = h_2 = \dots = h_n = h$ in (3) and

dividing (3) by h throughout, we get

$$\begin{bmatrix} 4 & 1 & 0 & \dots & 0 & 0 & \dots & 0 \\ 1 & 4 & 1 & \dots & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 & 4 & 1 & 0 \\ & & & & 0 & 1 & 4 & 1 \\ & & & & & 0 & 1 & 4 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_{n-1} \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ \vdots \\ d_{n-1} \end{bmatrix} \quad \text{--- (4)}$$

$$d_i = \frac{D_i}{h} = \frac{6(f_{i+1} - 2f_i + f_{i-1}))}{h^2}$$

Then find $S_i(x)$, by obtaining a_i from (4)

Eg Given,

i	0	1	2	3
x_i	1	2	3	4
$f(x_i)$	0.5	.3333	.25	.20

estimate $f(2.5)$, using cubic splines functions

Soln The points are equally spaced

$\therefore h_1 = h_2 = h_3 = 1$

$n = 4$, $\begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \end{bmatrix}$

$$d_1 = \frac{6(f_2 - 2f_1 + f_0)}{1} = \frac{6(.25 - 2(.333) + .5)}{1}$$

$$= .5004$$

$$d_2 = \frac{6(f_3 - 2f_2 + f_1)}{1} = \frac{6(.20 - 2(.25) + .3333)}{1}$$

$$= 0.1998$$

$$\begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} .5004 \\ .1998 \end{bmatrix}$$

$$\Rightarrow 4a_1 + a_2 = .5004$$

$$a_1 + 4a_2 = .1998$$

Solving, $a_1 = .1201$ & $a_2 = .0199$

$$\begin{aligned}
 S_1(x) &= \frac{a_0}{6} (h_1^2 u_1 - u_1^3) + \frac{a_1}{6} (u_0^3 - h_1^2 u_0) \\
 &\quad + \frac{1}{h_1} (f_1 u_0 - f_0 u_1) \quad (\text{by } \textcircled{1}) \\
 &= \frac{.1201}{6} [(x-1)^3 - (x-1)] + \left[\frac{.333(x-1)}{.5(x-2)} \right]
 \end{aligned}$$

$$0.5 \leq x \leq 1$$

To evaluate $f(2.5)$, we need to find $S_2(x)$, $\therefore$ by $\textcircled{1}$, we get

$$\begin{aligned}
 S_2(x) &= \frac{a_1}{6} (u_2 - u_2^3) + \frac{a_2}{6} (u_1^3 - u_1) + (f_2 u_1 - f_1 u_2) \\
 &= \frac{.1201}{6} [(x-3) - (x-3)^3] + \frac{.1998}{6} [(x-2)^3 - (x-2)] \\
 &\quad + [.25(x-2) - .333(x-3)]
 \end{aligned}$$

$$\Rightarrow S_2(2.5) = \underline{0.2829}$$